

Filament Routing Attachment

Design and Manufacture Report

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Printer	Ender-3 V3 SE
Material	PLA
CAD Tool	FreeCAD
Version	1.0

Part of an open-source engineering portfolio.

GitHub: <https://github.com/Bren10/Filament-Routing-Attachment>

1 Introduction

Problem I use multiple filaments for my Ender-3 V3 SE, often months go by between uses. Specific types of filament, such as PLA, PETG and nylon degrade as the filament absorbs moisture. To remove the moisture I use a Sunlu filament dryer along with my Ender 3. However, the filament heater positioned beside the printer requires routing filament from table height up and back down to the extruder inlet. The stock filament holder on the Ender 3 does not accommodate an elevated feed path, causing the filament to bind at the entry point or collide with larger prints. My current setup can be seen in Fig.1.



Figure 1: My Current Ender-3 V3 SE setup

This report documents the design, manufacture, and testing of a 3D printed filament guide attachment that mounts in place of the stock holder and provides a smooth, low-friction feed path for the elevated filament routing configuration.

2 Design

2.1 Constraints and Goals

The following constraints were identified prior to design:

- Must mount to the existing filament holder location with no permanent modification to the printer frame.
- Must accommodate filament [diameter, 1.75 mm] without excessive clearance that could cause buckling.
- Bend radius must be sufficient to prevent filament binding or fatigue cracking during continuous use.

- Should print without support material where possible to minimise post-processing.
- Must be set high enough so the filament line cannot collide with larger prints.
- Must be able to withstand the partial weight of the dryer, Approx 1kg (Sometimes filament gets tangled inside the dryer and pulls the dryer over.).
- Must be able to resist heat that is generated from the printing of parts on the Ender 3.

2.2 Design Concepts

There are two major considerations for the filament design that need to be considered. The interface with the Ender-3 V3 SE, and how the filament is to be guided. These considerations will be evaluated independently then combined together as the final design.

2.2.1 Printer Attachment Interface

Original Design One concept for the attachment of the Filament Guide would be to reuse the original handle that hold up the filament spool. This could be a simple and approved design that has been shown to work. The original attachment as well as the printed concept can be seen in Fig.2.



Figure 2: My Current Ender-3 V3 SE setup

Pro's	Con's
• Proven Design • Inner hole radius can be decreased to increase strength.	• Requires support • Required accurate tolerance between nut and attachment.

Bolted into Gantry Frame The second concept would be to bolt the designs directly into the Gantry stand, reusing the bolt holes that were used for the rack. This design would take advantage of the original printer interface and can be removed via unbolting.

2.3 Bend Radius Selection

A minium bend radius was assumed using the spool of a previous PLA filament.

2.4 Geometry and Mounting

[Describe the guide geometry and mounting approach here.]

2.5 Material Selection

I have

3 Manufacturing

3.1 Print Settings

Parameter	Value
Printer	[Model]
Material	[Material]
Layer Height	[e.g. 0.2 mm]
Infill	[e.g. 40%]
Perimeters	[e.g. 3]
Supports	[Yes / No]
Print Time	[approximate]

3.2 Post-Processing

[Describe any post-processing here, or note that none was required.]

4 Results

4.1 Version 1

[Describe v1 results here.]

4.2 Version 2 (if applicable)

[Describe v2 changes here, or remove this subsection if v1 was final.]

5 Conclusions

[Summarise whether the guide solved the original problem, any limitations of the current design, and potential future improvements.]

References

- [1] J. H. Ferziger, M. Perić, and R. L. Street, *Computational Methods for Fluid Dynamics*. Springer International Publishing, 2020. DOI: 10.1007/978-3-319-99693-6.